

IT5005 Artificial Intelligence

Tutorial 6.

Question 1

Consider the full Tic-Tac-Toe search space shown in Figure 1. The TicTac-Toe search space can actually be reduced by means of symmetry. This is done by eliminating those states which become identical with an earlier state after a symmetry operation (e.g. rotation). The following diagram shows a reduced state space for the first three levels with the player making the first move using "x" and the opponent making the next move with "o". Assume that the following heuristic evaluation function is used at each leaf node n :

$$Eval(n) = P(n) - O(n)$$

where $P(n)$ is the number of winning lines for the player while $O(n)$ is the number of winning lines for the opponent. A winning line for the player is a line (horizontal, vertical or diagonal) that either contains nothing or "x". For the opponent, it is either nothing or "o" in the winning line. Thus, for the leftmost leaf node in Figure 2, $Eval(n) = 6 - 5 = 1$.

- Use the minimax algorithm to determine the first move of the player, searching 2-ply deep search space shown in Figure 2.
- Assume that the "x" player will now make his second move after his opponent has placed an "o". Complete the following minimax tree in Figure 3 by filling the remaining blank boards at the leaf nodes. Compute the evaluation function for each of the filled leaf nodes and determine the second move of the "x" player (searching 2-ply deep).

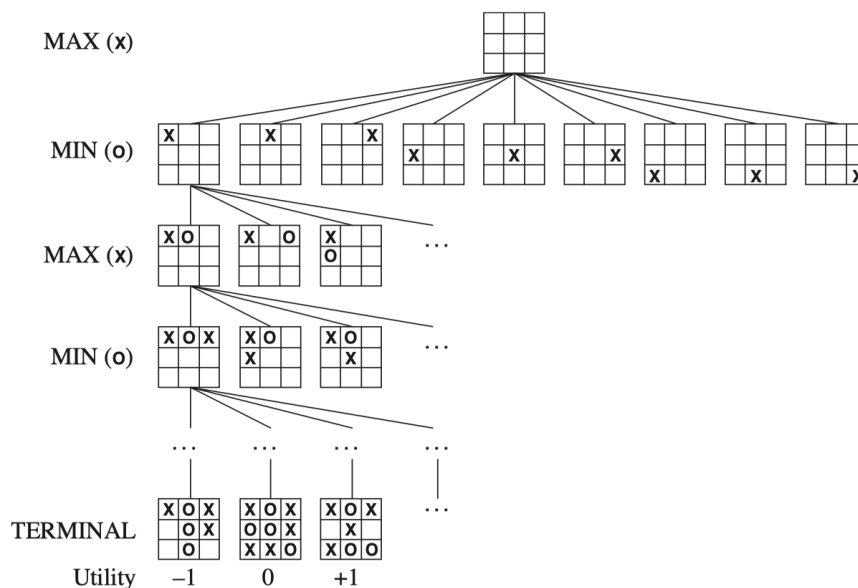


Figure 1: Full search space

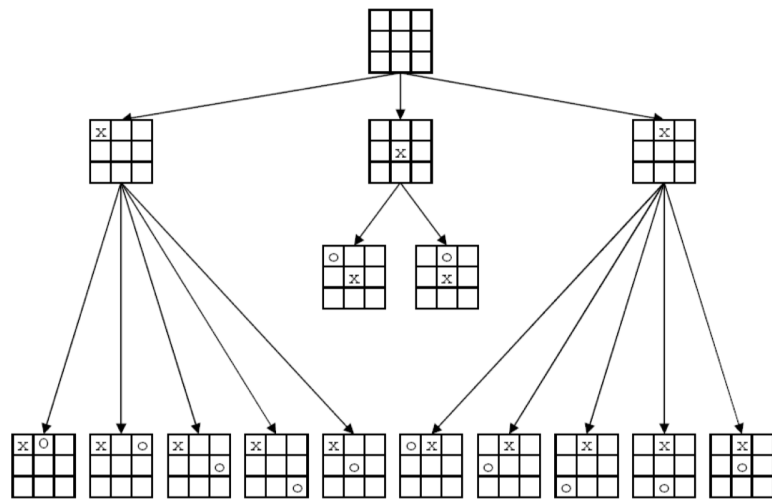


Figure 2: 2-ply deep search space

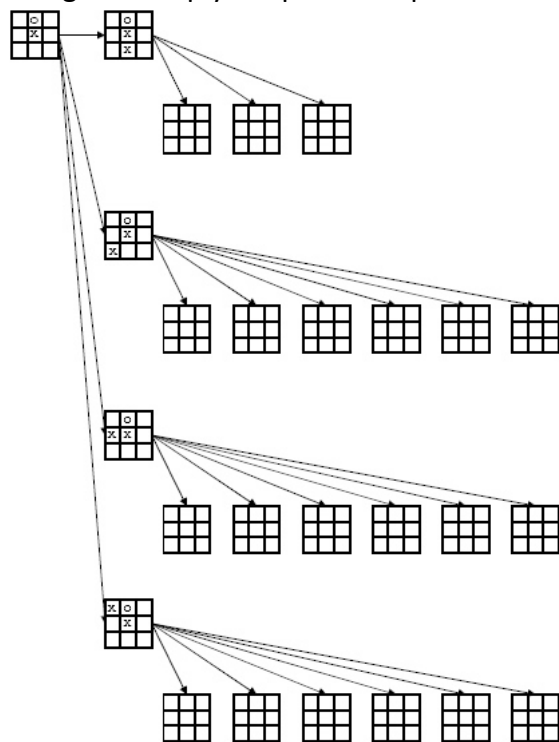


Figure 3: 3-ply deep search space

Question 2

Consider the following game: we have an attacker looking at three targets: t_1, t_2 and t_3 . A defender must choose which of the two targets it will guard; however, the attacker has an advantage: it can observe what the defender is doing before it chooses its move. If an attacker successfully attacks it receives a payoff of 1 and the defender gets a payoff of -1 .

- (a) Model this problem as a minimax search problem. Draw out the search tree. What is the defender's payoff in this game?
- (b) Can the defender do better by randomizing? What is the defender's optimal strategy? Prove your claim.

Question 3.

We have a 1-D Kohonen map with 5 units (neurons) is shown below.

0	1	2	3	4
(3, 9)	(17, 9)	(3, 23)	(42, 20)	(4, 21)

The format for each unit is:

<cell number> <2D vector containing current unit contents>
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We are given the following input:

$X = (18, 7)$

- a. Using an L2 norm (a "norm" is just another name for distance) defined as:

$$l_2(X, Y) = \sum_{i=0}^{n-1} (x_i - y_i)^2$$

Compute the following table showing the distance between X and each of the units:

Unit	Distance between X and unit
0	
1	
2	
3	
4	

Which is the best-matching unit?

- Explain the purpose of the neighborhood function in Kohonen maps.
- We have the following neighborhood function:

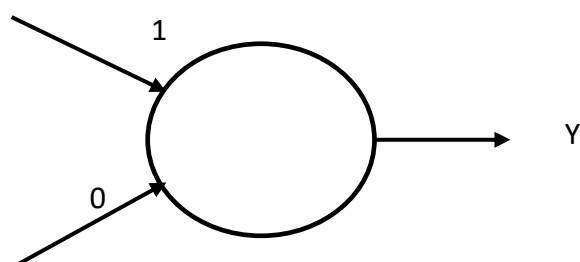
0.25	0.75	1	0.75	0.25
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Using a training rate $\alpha = 0.7$, update the Kohonen map based on your results in part a. Write out the new map below:

0	1	2	3	4
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Question 4.

- What is the purpose of the bias in Perceptrons? Show, using a suitable example, the importance if this bias.
- Does the value of the bias matter? Why or why not?
- We have a 1-input Perceptron shown below, with a bias input of 1:



Given the following data to be learnt:

X	Y
2	6
6	14.5
4	10.75

Complete the following table:

X	Y	Y _{out}	Error
2	6		
6	14.5		
4	10.75		

- d. Using the results in part c., apply the Perceptron learning law to update the weights, using a learning rate of $\alpha = 0.6$. Write the new weights on the diagrams below:

